

Metallic, elemental bismuth reacts with yellow fluorine gas to form a white solid of bismuth(V) fluoride according to the balanced equation above. How many grams of fluorine gas are needed to fully convert 836 g of bismuth in this reaction?

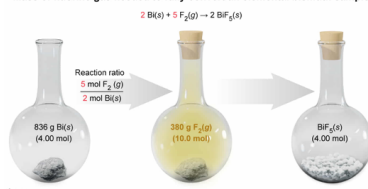
- ☐ A. 60.8 g [3%]
☐ B. 152 g [8%]
☐ C. 190 g [27%]
☒ D. 380 g [61%]

Correct

60% answered correctly

Explanation:

Mass of fluorine gas needed to fully convert an elemental bismuth sample



The **molar mass** of a compound is the amount of **mass contained in 1 mole** of the compound (ie, the grams per mole) and can be used to convert the mass of a compound to the equivalent number of moles. As such, the number of moles of Bi(s) in an 836 g sample can be found by dividing the mass of Bi(s) by its molar mass from the periodic table:

$$836 \text{ g Bi(s)} \times \frac{1 \text{ mol Bi(s)}}{209 \text{ g Bi(s)}} = 4.00 \text{ mol Bi(s)}$$

The number of moles of a chemical species is related to the number of moles of another chemical species in a reaction by the **mole ratios** of the **stoichiometric coefficients** from a **balanced reaction equation**. The given balanced reaction equation shows that 5 mol of F₂(g) are consumed for every 2 mol of Bi(s) that reacts. Applying this mole ratio, the number of moles of F₂(g) needed to fully convert the Bi(s) sample in the reaction is:

$$4.00 \text{ mol Bi(s)} \times \frac{5 \text{ mol F}_2(\text{g})}{2 \text{ mol Bi(s)}} = 10.0 \text{ mol F}_2(\text{g})$$

The mass of this number of moles of F₂(g) can be found by multiplying the moles of F₂(g) by its molar mass calculated from the periodic table (ie, molar mass F₂(g) = 2 × 19.0 g/mol = 38.0 g/mol).

$$10.0 \text{ mol F}_2(\text{g}) \times \frac{38.0 \text{ g F}_2(\text{g})}{1 \text{ mol F}_2(\text{g})} = 380 \text{ g F}_2(\text{g})$$

(Choice A) A mass of 60.8 g results if the mole ratio $\frac{5 \text{ mol F}_2(\text{g})}{2 \text{ mol Bi(s)}}$ is mistakenly inverted as $\frac{2 \text{ mol Bi(s)}}{5 \text{ mol F}_2(\text{g})}$ during the second step of the calculation.

(Choice B) A mass of 152 g is obtained by incorrectly assuming a 1:1 mole ratio for F₂(g):Bi(s) during the second step of the calculation.

(Choice C) A mass of 190 g is calculated if the molar mass of F (19.0 g/mol) is used instead of the molar mass of F₂(g) (38.0 g/mol) in the last step of the calculation.

Educational objective:

The molar mass (the amount of mass in 1 mole of a compound) permits conversions between a given amount of mass and the number of moles that it contains. Stoichiometric mole ratios from a balanced reaction equation relate the number of moles of a chemical species to the moles of another chemical species in a reaction.

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 Interactions of Chemical Substances